

Project Proposal

Scheduling with a Rules-Based Inference Engine Using Ontology

Problem Statement

Manually scheduling tasks involves a lot of unnecessary overhead for people with busy schedules. Offloading the need to schedule each task/event manually to an algorithm can be a great timesaver. Such an algorithm can also be used in corporate settings to improve efficiency and productivity, allowing employees to focus their time on more important and impactful matters.

This project will build a system that takes in a set of tasks (with appropriate attributes like deadlines, required energy levels for completion, etc), and outputs a suggested schedule for the week. Success for this project will look like a correctly functioning ontology and rule engine that correctly schedules inputted tasks without (at the bare minimum) violating constraints and (ideally) finding a schedule that maximizes tasks being done at times when the energy level is appropriate and in order of urgency and priority.

Approach Explanation

I will be tackling this problem by implementing a rules-based inference engine using ontology, and here's why:

- The human reasoning that goes into scheduling events and tasks goes beyond optimizing a single numerical value. Ontology makes representing the thought process and goals to be optimized easier and clearer.
- It's a transparent (in retrospect) algorithm (vs something like a genetic algorithm or reinforced learning algo). Every scheduling decision will have an explanation that comes from a rule that can be interpreted.

Scope

The main goal of this project, as stated, is to create a rules-based inference engine using ontology to schedule tasks "smartly" on a calendar. This project could easily expand out of scope for this class

Related Work

Usip et al. combined formal OWL ontology with first-order logic rules to provide an optimal timetable schedule. This work directly demonstrates how ontology can be used for scheduling, and how modeling real-world constraints with formal logic reasoning is an appropriate way to find an optimal schedule that abides by said constraints. The primary difference between the work of Usip et al. and my proposed project is that mine is focused on a personal scale, whereas their research is aimed at institutional scheduling. Their rules are more generalizable across users, while mine are specific to my case. Noticing this difference helped me better understand the scope of my project and got me thinking about how I could generalize it to work for others as well.

Citation

Usip, P. U., et al. (2019). A formal ontology-based framework towards an intelligent temporal scheduler. *Advances in Science, Technology and Engineering Systems Journal*, 4(2), 387–394.

Expected Deliverables

- Final Report (Motivation, Background, Study Design, Evaluation Criteria, Preliminary Results, Final Results, Conclusion)
- Final Presentation
- Dataset
- Ontology (OWL)
- Translation script (OWL, RDFLib)
- Rules-Based Inference Engine
- (Optional, if extra time) Alternative implementation (genetic algorithm, reinforced learning, etc)
- (Optional, if extra time) Google Calendar extension/integration using API